

Assignment 1 - CSE340

1. Explain the concept of the power wall with an example.
2. Calculate the spec ratio using the given table.

Description	Name	Instruction Count x 10 ⁹	CPI	Clock cycle time (seconds x 10 ⁻⁹)	Execution Time (seconds)	Reference Time (seconds)
Perl interpreter	perlbench	2684	0.42	0.556	627	1774
GNU C compiler	gcc	2322	0.67	0.556	863	3976
Route planning	mcf	1786	1.22	0.556	1215	4721
Discrete Event simulation - computer network	omnetpp	1107	0.82	0.556	507	1630
XML to HTML conversion via XSLT	xalancbmk	1314	0.75	0.556	549	1417
Video compression	x264	4488	0.32	0.556	813	1763

3. Suppose two custom computing chips, AlphaChip and BetaChip, use different architectural designs. The instructions they support can be divided into four classes according to their CPI (Class A, B, C, and D). AlphaChip has a clock rate of 2.7 GHz and the instruction classes have CPIs of 7, 2, 3, and 6 respectively, whereas BetaChip has a clock rate of 3.0 GHz and the instruction classes have CPIs of 5, 4, 2, and 1 respectively.

Now suppose a program has an instruction count of 1.0×10^6 , and the instructions are divided into classes as follows:

30% class A,

50% class B,

10% class C, and

10% class D.

Now answer the following questions:

- a. Calculate how many more clock cycles per instruction on average does the AlphaChip take compared to the BetaChip?
- b. Calculate the difference between the execution time in these two processors in milliseconds.
- c. If the program runs on a reference PC with an execution time of 120 ms, calculate the SPECRatio for the AlphaChip.